

Reducing Gulo 3000 shaft vibration

1 / Experimental basis and mathematical model

Four passive changes can cut modeled vibration by ≥30%

Gulo 3000 | Experimental model + design simulations | Powertrain Dynamics Group | ME 495

First prototype to investigate: a shaft-path damper that preserves modeled steady-speed response.

1 Measure and model the shaft

Motor switching excites torsional vibration. Goal: validate its source and identify four ≥30% solutions.

PHOTO TO TAKE — physical test rig

Take a clear front-side photo. Label the motor, flywheels 1 and 2, shaft, distal bearing and both tachometers.

CAD / SCHEMATIC TO ADD — two-inertia system

Capture a simplified CAD assembly or draw a clean schematic. Label Jc, J, K, Km, Bm, Bs, voltage input Vin and speed output ω.

Parameters come from measurements and mechanics

Parameter Identified value

Motor resistance Rm 2.495 Ω

Motor constant Km 0.0982 V·s/rad

Combined damping (Bm) 0.001429; 0.007808 N·m·s/rad

Effective shaft stiffness K 2.2029 N·m/rad (reduced)

Locked motor free decay

Motor mechanically uncoupled

3 sweeps + 28 time records. Free decay identifies damping; the uncoupled motor test estimates external loss.

2 The flexible model explains resonance

Resonance: 5.65 Hz measured / 5.65 Hz model

4–10 Hz fit error: 1.65 dB magnitude; 7.82° phase

Sensitivity separates attenuation from detuning

All 12 parameters were screened. Diameter strongly shifts the peak Rm, Km, Bm and Bs most strongly change its height.

3 Four quantified design alternatives

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Change Parameter Change % Power %

Resistance 4.50 Ω 32.2% 38.8%

Motor constant 0.093 V·s/rad 34.7% 45.2%

Shaft damping 0.038 N·m·s/rad 85.0% 32.2%

Motor damping 0.077 N·m·s/rad 77.6% 32.2%

Shaft damper: no local peak; gain assessed over 4–10 Hz.

All four reduce the 1-Hz response cost by ≥30%.

Prioritize the shaft damper. Its large damping increase needs hardware testing; the other options sacrifice steady-speed response.

Recommendation. Investigate a shaft-path damper with $B_s = 0.039 \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$. It reduces modeled switching ripple by 32.2% and preserves steady-speed gain. Four alternatives meet the nominal 30% target under the stated fixed-voltage test. These are model predictions; no modified hardware was tested.

Data and experimental procedures

The supplied September 10 records contain 3 distinct frequency sweeps (182 rows, 4–10 Hz) and 28 time histories. The copied/renamed exports are byte-identical to the originals and are not counted twice. The LabVIEW headers already express both tachometers in rad/s. Gain is distal-speed amplitude divided by motor-voltage amplitude; repeated complex responses are averaged in 0.05-Hz bins (44 bins). Phase is treated circularly before unwrapping. [2,3]

DC resistance uses the two stationary tests. Km is the slope of $V - R_m i$ versus steady motor speed across 14 records, with a free intercept. The 20-Hz recordings are retained as diagnostics rather than pooled as DC resistance. Locked-side free decays give $B_{tr} = B_s + B_b$ and $B_{tl} = B_s + B_m$. For the two coastdowns, the motor was physically uncoupled while both flywheels remained connected by the shaft (confirmed by the team).

Equations and parameter provenance

States: shaft twist $\delta = \theta_1 - \theta_2$, motor speed ω_1 , distal speed ω_2 and armature current i . The model uses $J_c = J_m + J_f$ and the same J_f for both flywheels. [2]

$$L_m di/dt = V_{in} - R_m i - K_m \omega_1$$

$$J_c d\omega_1/dt = K_m i - B_m \omega_1 - B_s(\omega_1 - \omega_2) - K\delta$$

$$J_f d\omega_2/dt = B_s(\omega_1 - \omega_2) + K\delta - B_b \omega_2; d\delta/dt = \omega_1 - \omega_2$$

The frequency model $H(s) = \Omega_2(s)/V_{in}(s)$ neglects L_m ; switching simulations retain it. $J_f = \pi r D_f^4 L_f / 32$ and $K_{geometry} = \pi G_s D_s^4 / (32 L_s)$. The complete third-order polynomial and an independent state-equation check are in the MATLAB functions.

Parameter	Value	Basis
Rm; Km	2.4947 Ω; 0.09821 V·s/rad	Stationary tests; back-EMF regression
Jf; Jm	0.003406; 4.23e-05 kg·m ²	Given geometry/material; given rotor inertia
Kgeometry → Keffective	2.4062 → 2.2029 N·m/rad	Geometry; only parameter fitted to sweeps
Btr; Btl	0.001429; 0.007808 N·m·s/rad	Mean of two free decays per locked side
Bb; Bs; Bm	0.0001572; 0.001272; 0.006536 N·m·s/rad	Coastdown and subtraction of damping sums
Lm; shaft geometry	0.002 H; Ds = 3.18 mm; Ls = 305 mm	Given course parameters [2]

Damping extraction and its limits

Free-decay extrema are fitted as $\ln|\omega_{peak}| = c - \alpha t$; $B = 2J\alpha$, $\omega_n = \sqrt{[(2\pi f_d)^2 + \alpha^2]}$ and $\zeta = \alpha/\omega_n$. Extrema above 15% of the initial amplitude, within 8 s, are retained. The coast fit uses 4.0–19.5 s after disengagement and $J = 2J_f$, giving $B_b \approx 2J_f\alpha$. Then $B_s = B_{tr} - B_b$ and $B_m = B_{tl} - B_s$. All three are effective coefficients of this assembly, not independently isolated material properties.

The motor regression has $R^2 = 0.99981$ and intercept 0.135 V. The intercept and effective resistance absorb offsets/brush effects. Coastdown is slightly better fit by a linear Coulomb model than an exponential model; an equivalent viscous model is retained for frequency-domain design. This approximation limits extrapolation.

2 / Validation and design screening

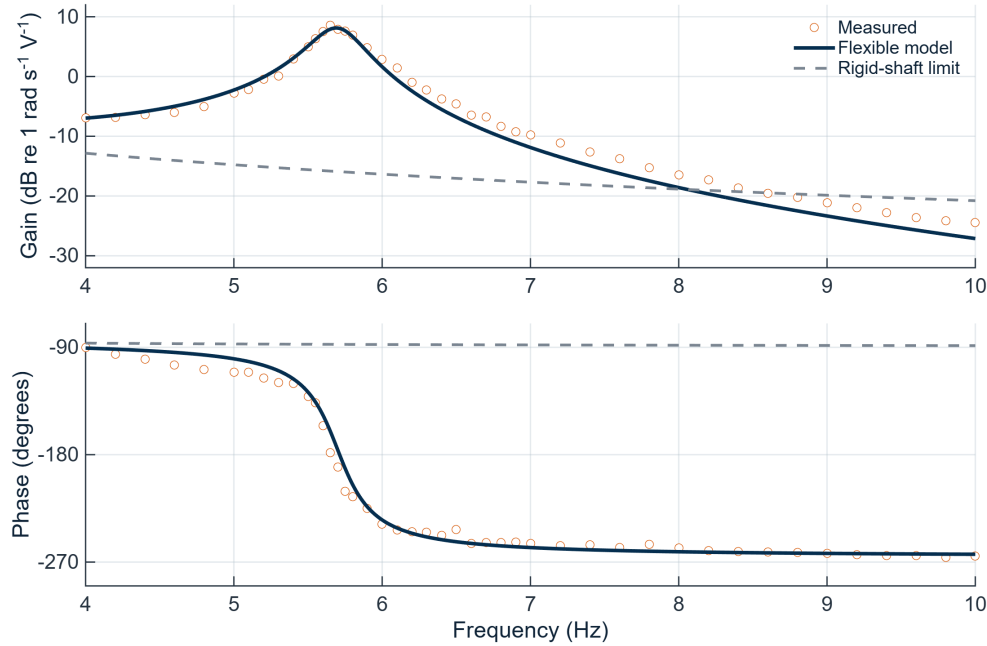


Figure 1. Measured response, flexible model and rigid-shaft limit. A rigid shaft enforces $\omega_1 = \omega_2$ and removes the torsional resonance; setting $B_s = 0$ does not make a shaft rigid.

Check	Magnitude / frequency result	Phase / interpretation
Measured vs modeled resonance	5.650 vs 5.690 Hz	Peak gain 2.687 vs 2.547 rad/(s·V)
Before → after K refinement	2.46 → 1.65 dB RMS	28.38° → 7.02° RMS
Leave one sweep out	0.89–1.83 dB RMS	5.03–17.47° RMS; shared apparatus
Broadband last-test diagnostic	2.69 dB RMS	8.98° RMS; coherence ≥ 0.8

K is refined from geometry by fitting mean squared log-magnitude and wrapped-phase residuals. It is an effective stiffness correction, not a new measured shear modulus. Near resonance (5.2–6.2 Hz), magnitude RMS error is 0.79 dB. Including the given inductance changes predictions by at most 0.031 dB and 2.88° in the measured band.

The broadband record is withheld from fitting: an H1 spectral estimate uses nine 2-s Hann windows with 50% overlap and 0.5-Hz resolution. Nine 4–10 Hz bins meet coherence ≥ 0.8 . Its coarse resolution cannot resolve the narrow peak accurately. The three sweeps share setup and nearly constant excitation (~1.72–1.76 V amplitude); no descending sweep or systematic amplitude sweep establishes broad linearity. The model supports comparative design near the tested range, with residual discrepancies retained.

Sensitivity identifies four attenuation directions

Course cost: $J = \int |H(j\omega)| d\omega$ over $\omega r \pm \pi$ (a 1-Hz-wide band); integrate linear magnitude, not dB. Normalized sensitivity is $S_p = (\Delta J/J)/(\Delta p/p)$, estimated with symmetric $\pm 1\%$ perturbations. Record both the original-band cost and the cost following the new resonance, plus peak-height sensitivity. Recompute linked parameters after each perturbation. [2]

Parameter	Original-band cost S	Tracked-peak S	Interpretation
Rm	-0.869	-0.757	Attenuation candidate
Km	+0.737	+0.513	Attenuation candidate
Bm	-0.222	-0.412	Attenuation candidate
Bs	-0.176	-0.326	Attenuation candidate
Df	-3.201	-0.048	Primarily moves resonance
Ds	+1.396	-0.017	Primarily moves resonance

The four largest absolute peak-height sensitivities are Rm, Km, Bm and Bs. All 12 raw parameters were screened; Jm, Lf, pf, Ls, Gs and Bb are also included in sensitivity.csv. Changing Df changes both flywheel inertias as Df^4 ; Ds changes stiffness as Ds^4 . Geometry can move the resonance out of a fixed cost band while leaving its height nearly unchanged, so it is not selected on fixed-band improvement alone.

3 / Four design alternatives and recommendation

Targets below include a modest parameter margin beyond the minimum joint solution and are rounded outward to two significant digits. Each nominal design reduces both the frequency-response criterion and the defined switching ripple by at least 30%. These are component-level targets for prototype evaluation, not validated vehicle designs.

Parameter change	Gain ↓	1-Hz cost ↓	Pulse ↓	Speed gain retained
Rm: 2.495 → 4.10 Ω	32.7%	35.9%	38.5%	71.0%
Km: 0.0982 → 0.053 V·s/rad	34.7%	40.5%	45.0%	72.9%
Bs: 0.00127 → 0.039 N·m·s/rad	85.9%	82.6%	32.2%	100.0%
Bm: 0.00654 → 0.077 N·m·s/rad	77.6%	65.9%	32.2%	13.0%

For Bs the local peak is eliminated: “gain” is the maximum across 4–10 Hz, and the 1-Hz cost stays centered on the baseline resonance. For other designs, both metrics track their shifted local peak. Broadband low-frequency commanded rotation is not relabeled as torsional resonance.

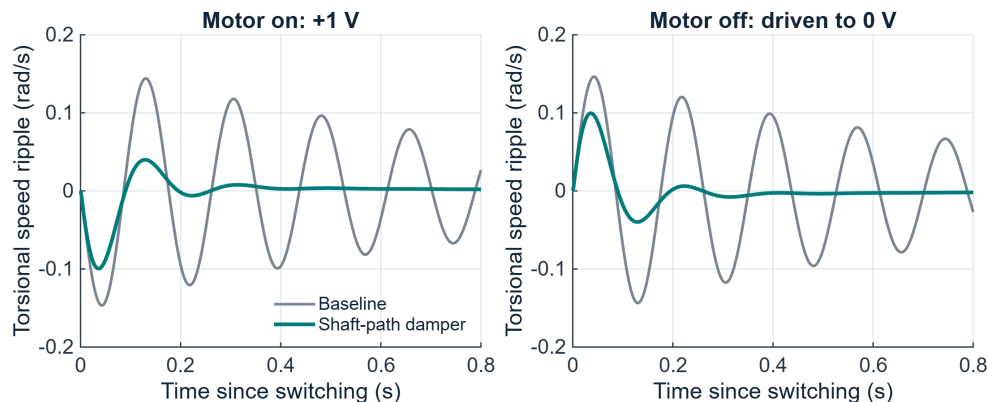


Figure 2. Baseline and shaft-damper switching responses. Exact matrix-exponential simulation retains motor inductance; +1 V for 4 s, then a driven 0-V source for 4 s. Initial state is rest. Plots show the first 0.8 s after each switch.

Torsional ripple is $r = \omega 2 - (Jc\omega 1 + Jf\omega 2)/(Jc + Jf)$. The criterion is $\max|r|$ over the full pulse: baseline 0.1467 rad/s. It measures relative torsional motion, not total wheel speed or passenger acceleration. An open-circuit turn-off variant is also saved. The design gain/cost thresholds alone would need much smaller damping changes (see `resonance_only_designs.csv`) but would not achieve the pulse target.

Practical comparison and parameter linkage

1. Shaft-path damper — preferred concept. Add relative-motion dissipation across the shaft/coupling path, targeting Bs = 0.039 N·m·s/rad. This is about 30.7 times the baseline coefficient and retains 100% of modeled steady-speed gain. Specify the damper by its measured torque–relative-speed behavior. Added stiffness, inertia, heat and packaging must be included in a new prototype model.
2. Series resistance. Target total Rm = 4.10 Ω (about 1.61 Ω added). It is a simple passive concept, but produces I^2R heat and retains only 71% of steady-speed gain. The input is then defined upstream of the added resistor; do not use downstream motor-terminal voltage as the same transfer-function input.
3. Lower-Km motor. Target Km = 0.053 V·s/rad (also N·m/A in SI). The idealized study holds Rm, Jm and damping fixed; real motor replacement changes these together. Refit all coupled properties and verify torque/current requirements. Modeled steady-speed gain retained: 73%.
4. Motor-side damper. Target Bm = 0.077 N·m·s/rad. This resists absolute motor speed and causes continuous dissipation. Only 13% of steady-speed gain remains; this makes it the least attractive of the four despite meeting the fixed-voltage target.

Fixed input is a material limitation. If voltage is rescaled to restore equal final speed, pulse reductions are approximately Rm: 13.5%, Km: 24.6%, Bs: 32.2%, Bm: -420.1%. A negative reduction means worse ripple. These rescaled simulations are diagnostic and are not recommended drive voltages. Eight combinations of the repeated damping tests are checked in `replicate_robustness.csv`; they are not statistical confidence intervals.

References and reproduction

[1] Meyhofer & Reddy, Task Letter Lab 1, Sept. 4, 2026. [2] ME 495 Lab 1_1, pp. 12–25; Lab 1_2, pp. 4, 19–36, 43. [3] Flexible Shaft — Student Guide, pp. 20–27. Poster format follows Dr. Royston, F26 ME495 Posters-1, pp. 20–28, 48, 54–55. All numerical results: supplied Data/ exports; run MATLAB/run_all.m. Tables, diagnostic plots and verification results are in Analysis/; publication figures are in Figures/.